

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering

Course Code & Title: CS8451 & Design and Analysis of Algorithms

Name of the Faculty member: Ms.P.Jothi Thilaga, AP/CSE

Date: 27.01.2020

Innovative Practice: Think-Pair-Share

Topic: Closest-Pair and Convex Hull problem

Type of Learning:

Active Learning

Learning Objectives:

O1: To enhance the students to gain knowledge in the concept of Closest-Pair and Convex Hull problem.

Description:

Think-Pair-Share is a collaborative, active learning strategy. Initially the students were asked to think about the given topic individually without discussing with other in the class (Think). Next, pair up or make a group with others in the class and discuss their points with them and also listen their views (Pair). Finally share points with the entire class (Share).

Uses of Think-Pair-Share:

- It helps the students to think individually about a topic or answer to a given question.
- It makes the students to share ideas with classmates and builds oral communication skills.
- It helps to focus attention and engage students in learning.

Justification for chosen the topic:

Closest-Pair and Convex Hull problem is one of the important topics, repeatedly asked in university questions. This activity makes the students to get a sound knowledge in this concept. Students think individually about a topic and share their ideas with their classmates that improve creative thinking and oral communication skills.

Implementation of Think-Pair-Share :

- Students were asked to think about the topic “Closest-Pair and Convex Hull problem” and make a note on it individually for five minutes.
- Next the students are grouped with other students (4 in a group) in the class to share their ideas about the given topic and also listen to others view on that topic for the next five minutes.
- Finally five volunteers S.Hemalatha, S.Mohan Babu, S.Sneha, P.Shanthi Priya and R.Sriram from the various groups shared about the concepts of convex hull and closest-pair problem to the entire class.



Outcomes:

- This activity actually increases attentiveness in the class and also improves the students' skill of critical analysis of the topics.
- This activity makes the students to understand the topic clearly.

Relevance to POs

Objective	PO1	PO2	PO3	PO4	PO5	PO9	PO10
O1	3	3	2	2	2	3	2

Reflective Report:

Identified Problems

- Some students are hesitated and lack to share the concepts that they learnt in the class.

Initiatives to address the problems

- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.
- Addressed the students that the activity is not a kind of evaluation methodology to assess.

Post-implementation

- Students were actively participated in this activity.
- From this activity, the students can get more clarity in the particular topic by discussing and sharing their views with the other students in the class.
- The graphical representations of the asymptotic notation were again explained for their better understanding.

References:

- <https://serc.carleton.edu/sp/library/interactive/tpsware.html>
- https://www.ritrjpm.ac.in/images/computer-science/6_IT6801_TPS.pdf

Faculty In-charge

Ms.P.Jothi Thilaga, AP/CSE

HOD-CSE

Dr.K.Vijayalakshmi